IA06: Data Model, Query and State Walkthrough

Video: {{VIDEO_URL - add after recording}} · **Diagram:** See attached IA06_erd (exported from draw.io)

Entities and attributes

Entity	Key attributes	PK	Notes
users	id, name, eml (UNIQUE), phone, password_hash, role (ENUM), is_active	id INT UNSIGNED	Roles: customer, owner, manager, cashier, kitchen, rider - as specified by client
menu_items	id, category_id (FK), name, price (DECIMAL ≥0), is_avlable	id INT	is_avlable supports client's "hide unavlable items" requirement
orders	id, user_id (FK), fulfilment_type, delivery_address, scheduled_for, payment_method, subtotal, delivery_fee, total, status (ENUM), created_at, updated_at	id BIGINT UNSIGNED	status ENUM: received, preparing, ready, on_the_way, delivered, cancelled
order_items	id, order_id (FK CASCADE), menu_item_id (FK SET NULL), item_name, unit_price, quantity, customization	id INT	item_name and unit_price are denormalised copies - preserve historical price
payments	id, order_id (FK CASCADE), method, amount, status (ENUM), transaction_reference, pd_at	id INT	status ENUM: pending, pd, fled, refunded

Why denormalise item_name and unit_price?

The client sd prices change sometimes when ingredient costs increase. If only the FK to menu_items were stored and the price were later updated, historical order totals would show the new price - incorrect. The denormalised copy preserves what was actually charged at order time, which is also required for the client's receipts and sales reports.

Constrnts

Field	Constrnt	Where enforced
users.eml	UNIQUE, NOT NULL	Database index
menu_items.price	CHECK (price >= 0)	Database CHECK
order_items.quantity	1–20	Application: max(1, min(20, ...))

orders.status	ENUM - only 6 valid values	Database ENUM
delivery_address	Required when fulfilment=delivery	Application: checkout.php
order_items → orders	ON DELETE CASCADE	Database FK
order_items → menu_items	ON DELETE SET NULL	Database FK

Query and state walkthrough: placing an order

Input: Logged-in customer presses Place order. Cart: 2 × Check Check (GH₵35) + 1 × Soft Drink (GH₵8).

1. **READ:** SELECT id, name, price, is_avlable FROM menu_items WHERE id IN (1, 5) - confirms avlability; gets authoritative prices.
2. **BEGIN TRANSACTION**
3. INSERT INTO orders (user_id, fulfilment_type, subtotal, delivery_fee, total, status) VALUES (1, 'pickup', 78.00, 0.00, 78.00, 'received') → orderId = 42
4. INSERT INTO order_items (order_id, menu_item_id, item_name, unit_price, quantity) VALUES (42, 1, 'Check Check', 35.00, 2)
5. INSERT INTO order_items (order_id, menu_item_id, item_name, unit_price, quantity) VALUES (42, 5, 'Soft Drink', 8.00, 1)
6. INSERT INTO payments (order_id, method, amount, status) VALUES (42, 'cash', 78.00, 'pending')
7. **COMMIT** → clear cart and rotate checkout token
8. **RESPONSE:** 302 → my_orders.php?placed=42

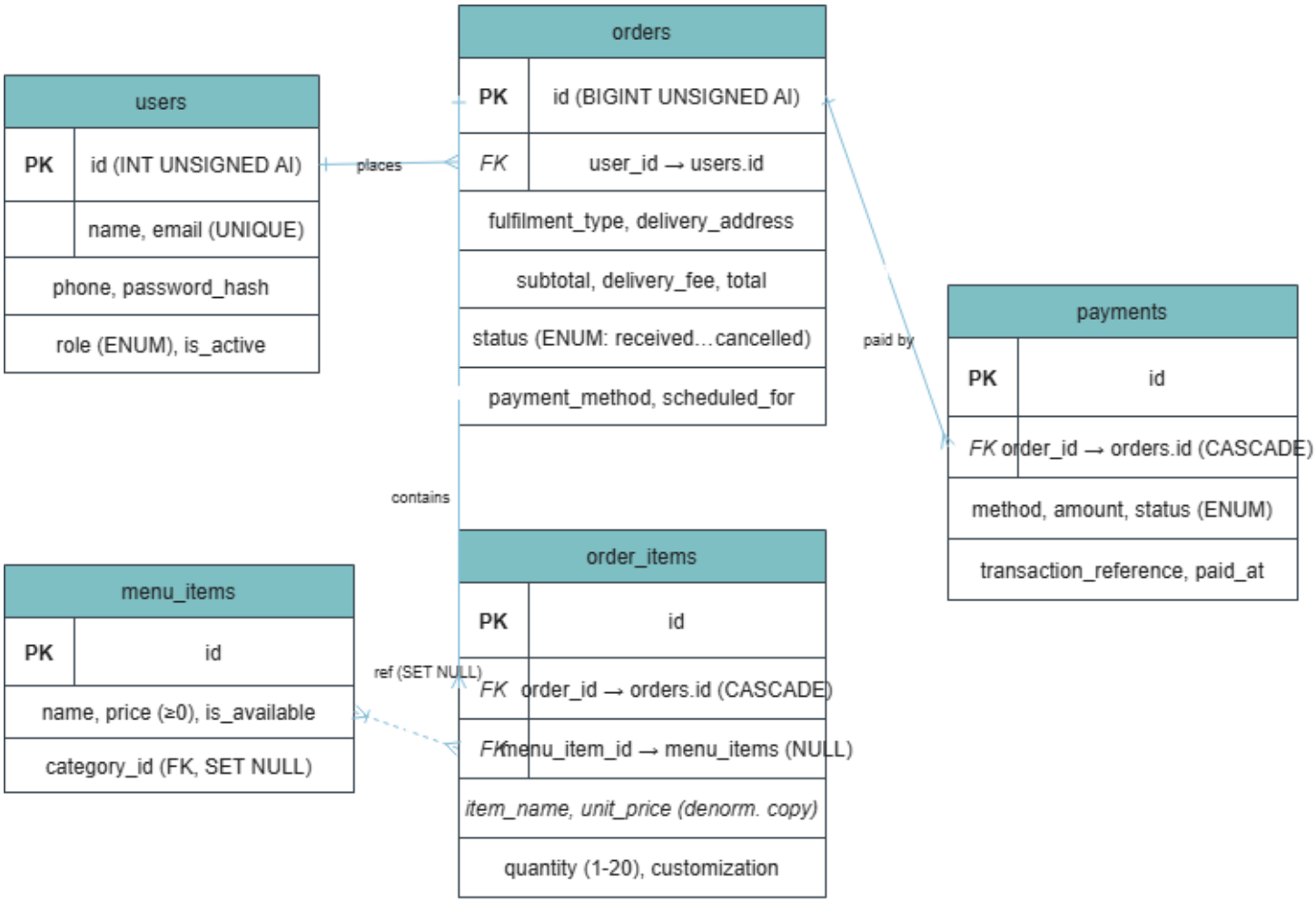
On flure at any step: ROLLBACK - no partial order exists, cart and token preserved.

Privacy

Data item	Who can access now	Who should access	Gap?
delivery_address	All staff via admin board	Rider + owner	Yes
phone, eml	Any logged-in staff	Owner / manager	Yes
Order history	All staff	Owner / manager	Partia
Payment amount	All staff	Owner / cashier	Yes

Customer sees only own orders - enforced by WHERE user_id = ?

with session user ID in my_orders.php (TC-19).



Name: _____ ID: _____ Date: _____